Status	Finished
Started	Sunday, 11 October 2026, 10:42 AM
Completed	Sunday, 11 October 2026, 10:45 AM
Duration	3 mins 28 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Merge Sort
- b. Binary Search
- c. Bubble Sort
- d. Linear Search

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It avoids the need for recursion
- b. It simplifies the sorting process by using only one pass
- c. It allows for efficient parallel processing and sorting of data
- d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **3**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It only works with integer arrays
- b. It requires external libraries
- c. It sorts data out of the box efficiently
- d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently

Question **4**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Merge Sort
- b. Bubble Sort
- c. Insertion Sort
- d. Quick Sort

The correct answer is: Merge Sort

Question **5**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Makes data harder to manage
- b. Makes code execution slower
- c. Provides a basis for other algorithms to work efficiently
- d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **6**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sort()
- b. sorted()
- c. order()
- d. arrange()

The correct answer is: sorted()

Question **7**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is difficult to implement
b. Bubble Sort cannot handle duplicate elements
c. Bubble Sort is less efficient for large lists
d. Bubble Sort does not guarantee sorted order

The correct answer is: Bubble Sort is less efficient for large lists

Question **8**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Backtracking
b. Divide and conquer
c. Greedy method
d. Dynamic programming

The correct answer is:
Divide and conquer

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It divides the list into two halves, sorts each half, and then merges them
b. It compares adjacent elements and swaps them if necessary
c. It repeatedly finds the minimum element and moves it to the sorted part of the list
d. It builds a sorted array one element at a time

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **10**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Merging
b. Complexity
c. Searching
d. Sorting

The correct answer is:
Complexity

Question **11**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Searching
b. Sorting
c. Rearranging
d. Merging

The correct answer is: Sorting

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Merge Sort
b. Bubble Sort
c. Insertion Sort
d. Quick Sort

The correct answer is: Bubble Sort

Question **13**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Heap sort
b. Quick sort
c. Selection sort
d. Merge sort

The correct answer is:
Merge sort

Question **14**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Being the most efficient sorting algorithm
b. Using the divide-and-conquer approach
c. Bubbling up the largest element to its correct position with each pass
d. Sorting data in a non-sequential manner

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Merging two datasets
b. Arranging data in a specific order
c. Removing duplicates from a list
d. Finding a specific element in a list

The correct answer is: Arranging data in a specific order